PYTHONS OPERATOR

In [ ]:
```
## operator
1- ARITHMETIC OPERATOR ( + , -, *, /, %, %%, **, ^)
2- ASSIGNMEN OPERATOR ( =, +=, -=, *=, /=, //=)
3- RELATIONAL OPERATOR ( <, >, ==, !=, >=, <=)
4- LOGICAL OPERATOR  ( AND, OR, NOT )
5- UNARY OPERATOR  ( " - " [Negattion] )
```

(1) ARITHMETIC OPERATOR

In [1]:
```
x1, y1 = 10, 5
```

In [2]:
```
#x1 ^ y1
```

In [3]:
```
x1  + y1
```

Out[3]:
```
15
```

In [4]:
```
x1 - y1
```

Out[4]:
```
5
```

In [5]:
```
x1 * y1
```

Out[5]:
```
50
```

In [6]:
```
x1 / y1
```

Out[6]:
```
2.0
```

In [7]:
```
x1 // y1
```

Out[7]:
```
2
```

In [8]:
```
x1 % y1
```

Out[8]:
```
0
```

In [9]:
```
x1 ** y1
```

Out[9]:
```
100000
```

In [10]:
```
x2 = 3
y2 = 2
x2 ** y2
```

Out[10]: 9

(2) ASSIGNMEN OPERATOR

In [11]:
```
x = 2
```

In [12]:
```
x = x + 2 # if you want to increment by 2
```

In [13]:
```
x
```

Out[13]: 4

In [14]:
```
x += 2
x
```

Out[14]: 6

In [15]:
```
x += 2
x
```

Out[15]: 8

In [17]:
```
x *= 2
```

In [18]:
```
x
```

Out[18]: 32

In [19]:
```
x -= 2
x
```

Out[19]: 30

In [20]:
```
x /= 2
x
```

Out[20]: 15.0

In [21]:
```
x //= 2
x
```

Out[21]: 7.0

In [22]:
```
a, b = 5,6 # you can assigned variable in one line as well
```

In [23]:
```
a
```

Out[23]: 5

In [24]:
```
b
```

Out[24]: 6

(3) RELATIONAL OPERATOR

Relational Operator use usally used for comparing

In [25]:
```
a < b
```

Out[25]: True

In [26]:
```
a > b
```

Out[26]: False

In [27]:
```
# a = b # we cannot use = operator that means it is assigning
```

In [28]:
```
a == b
```

Out[28]: False

In [29]:
```
a != b
```

Out[29]: True

In [30]:
```
# hear if i change b = 6
b = 5
```

In [31]:
```
a == b
```

Out[31]: True

In [32]:
```
a
```

Out[32]: 5

In [33]:
```
b
```

Out[33]: 5

In [34]:
```
a >= b
```

Out[34]: True

In [35]:
```
a <= b
```

Out[35]: True

In [36]:
```
a < b
```

Out[36]: False

In [37]:
```
a > b
```

Out[37]: False

In [38]:
```
b = 7
```

In [39]:
```
a != b
```

Out[39]: True

(4) LOGICAL OPERATOR

Three important parts of logical operators is ---> 1. AND 2. OR 3. NOT Let's underdtand the truth table :- IN Truth Table its represent with (True --> 1), (False --> 0)

In [41]:
```
a = 5
b = 4
```

In [42]:
```
a < 8 and b < 5 #refer to the truth table
```

Out[42]: True

In [43]:
```
a < 8 and b < 2
```

Out[43]: False

In [44]:
```
a < 8 or b < 2
```

Out[44]: True

In [45]:
```
a>8 or b<2
```

Out[45]: False

In [46]:
```
x = False
x
```

Out[46]: False

In [47]:
```
not x  # you can reverse the operation
```

Out[47]: True

(5) UNARY OPERATOR

* Unary Means 1 || Binary means 2 * Here we apply unary ,minus opratoe(-) on the operand n the value of m becomes -7, which indicate it as a negative value

In [48]:
```
n = 7 #negattion
n
```

Out[48]: 7

In [49]:
```
m = -(n)
m
```

Out[49]: -7

In [50]:
```
n
```

Out[50]: 7

In [51]:
```
-n
```

Out[51]: -7

In [53]:
```
u = 5
u
```

Out[53]: 5

In [55]:
```
u =-4
u
```

Out[55]: -4

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